

Synthesizability classification of generated structures

uvsib

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1 What it is

A MainWorkChain stage that classifies **every generated structure** of a composition by synthesizability, using three independent views (the three the literature splits into). It runs after the phase diagram is built and before the surface/adsorbate stages, so the 0 K hull it needs already exists.

It is post-processing only — no remote jobs. It reads the generated structures and their MLIP energies from the database, builds the hull, scores each structure, and writes the scores back.

2 Where it sits

MainWorkChain

```
pd_ml (generates + relaxes structures, builds hull)
pd_verification
--> synthesizability <-- this stage (workchains/main.py)
sqs
surface_builder
adsorbates
nano_generator
```

Gated like the other stages: skipped for the sqs and nano paths, skipped if `step_status["synthesizability"] == "Done"`, and globally toggled by `settings.SYNTH_ENABLED (input.yaml: synthesizability.enabled)`.

3 The three classifiers

All three live in `workchains/_synth_classifiers.py` (pure pymatgen/numpy, unit tested); the AiiDA wrapper is `workchains/synthesizability.py`.

View	Question	Signal
thermo	Is it thermodynamically accessible?	energy above the 0 K hull vs a metastability window
reaction	Will it form rather than its competitors?	formation energy from elemental precursors + the decomposition products it would form instead + polymorph selectivity at its composition
pu	Does it look like things that get made?	a positive-unlabeled learned score (loaded model, or a transparent surrogate)

3.1 1. thermo — thermodynamic accessibility

`e_above_hull` from `PhaseDiagram.get_e_above_hull`. Labels: product (on the hull, $\leq e_{\text{hull_tol}}$), metastable (within `meta_window` above), or inaccessible. Score = $\exp(-e_{\text{above_hull}} / \text{thermo_scale})$.

The window matters: metals/alloys sit at the tight end of the Sun/Ceder metastability scale, so `meta_window` defaults to 0.05 eV/atom — and the same caveat the literature flags applies: polymorph gaps can be below both MLIP and DFT-functional noise, so near-hull calls are exactly where a DFT check is worth its cost.

3.2 2. reaction — precursor → product selectivity

`PhaseDiagram.get_form_energy_per_atom` (reaction energy from the elemental precursors) and `get_decomp_and_e_above_hull` (the competing phases the composition decomposes into — the “products” that form instead). Plus a **polymorph gap**: the energy distance to the best generated structure at the same composition (0 $\Rightarrow$ this is the preferred polymorph). Labels: selective_product / competes / outcompeted.

E.g. an off-hull Cu_2Au is reported as decomposing into 0.667 Cu_3Au + 0.333 CuAu — the actual products, not just a number.

3.3 3. pu — positive-unlabeled synthesizability

Featurizes the composition (fraction-weighted mean/std of element properties: Z, electronegativity, group, row, mass, radius, Mendeleev number, + `n_elements`; matminer-free). If a pickled model with `predict_proba` is configured (`pu_model_path`), its probability is used. Otherwise a **transparent surrogate** — $\text{sigmoid}(2 - 18 \cdot e_{\text{above_hull}} - 0.4 \cdot (n_{\text{elements}} - 2))$, flagged model: `surrogate(ehull+complexity)` — so the stage runs end-to-end without a trained model. Drop a real PU classifier (SynthNN / CSPML / SynCoTrain-style, fit on the same featurizer over ICSD-positives) at `pu_model_path` to replace it.

3.4 Combined verdict

Weighted average of the three scores (weights, default equal), with a hard rule: an on-hull, selectively-preferred phase is synthesizable regardless. Labels by `synth_cut` / `maybe_cut`: synthesizable / maybe / unlikely.

4 Output

Per structure, the full record (thermo, reaction, pu, combined) is merged into that generated DB-StructureVersion's attributes JSONB under "synthesizability". A run-level summary (counts per label + the ranked synthesizable candidates) is stored on the DBComposition row's attributes and emitted as the workchain's synthesizability Dict output.

5 Configuration (input.yaml)

```
synthesizability:
  enabled: true
  ehull_tol: 0.005      # eV/atom; on-hull tolerance
  meta_window: 0.05    # eV/atom; metastable window above the hull
  thermo_scale: 0.04   # eV/atom; accessibility score decay
  complexity_penalty: 0.4 # PU surrogate penalty per element beyond binary
  synth_cut: 0.66      # combined -> "synthesizable"
  maybe_cut: 0.33      # combined -> "maybe"
  # weights: [1.0, 1.0, 1.0]      # thermo, reaction, pu
  # pu_model_path: /path/to/pu.pkl # trained PU model; omit -> surrogate
```

6 Scope note

Per the project scoping decision, the target space is metallic systems (alloys/intermetallics), not oxides/carbides — which is also where the hull, the metastability window, and the MLIP energies feeding this stage are most trustworthy. The honest limit is that all three views are **0 K energetics**: true synthesis-pathway/selectivity needs finite-T free energies (entropy, temperature) the potential can supply via phonons/MD but this stage does not yet compute.

7 Finite-temperature extension (phonons → Gibbs hull)

By default the three views score on the **0 K** hull. When synthesizability.finite_T.enabled is set, the stage first computes phonon free energies and scores on the **Gibbs hull at the synthesis temperature** — so vibrational entropy can flip which competing phase is the product.

Sub-step (per the user's spec):

```
near-hull candidates + hull vertices
-> PhononWorkChain (codes/phonon + codes/files/phonon.py)
    tight relax -> phonopy finite displacements -> MLIP forces
    -> force constants -> harmonic  $F_{\text{vib}}(T)$  -> QHA over a few volumes ->  $G(T)$ 
-> per-atom free-energy correction  $G(T) - E_0$  at the target T
-> finite-T (Gibbs) convex hull
-> classify on the finite-T hull
```

Only structures within ehull_window of the 0 K hull (plus the hull vertices) get phonons — far-above-hull phases won't be rescued by entropy, and phonons are expensive. Corrections are mapped back by structure UUID; the finite-T hull is rebuilt from all entries shifted by their $F_{\text{vib}}(T)$.

Hard requirement — conservative model. Phonons are second derivatives of the PES, so forces must be the analytic gradient of one energy. Use **MACE or MatterSim**; direct-force heads (eqV2/UMA, ORB) give asymmetric force constants and spurious imaginary modes and must not be used (finite_T.model, default MACE). Tight pre-relaxation (fmax 1e-3) avoids spurious imaginary modes; QHA captures the dominant thermal-expansion/Gibbs term without full anharmonic (TDEP/SCPH) treatment for most metals.

```

synthesizability:
  finite_T:
    enabled: false
    temperature: 1000.0      # K, the synthesis-temperature hull
    model: MACE              # conservative MLIP ONLY
    ehull_window: 0.10      # eV/atom; phonons only for near-hull structures
    displacement: 0.01      # Å; do a convergence check per family
    volume_scales: [0.97, 0.99, 1.0, 1.01, 1.03]

```

Two honest limits (literature, 2024–26): (1) **absolute** vibrational free energies from foundation MLIPs carry a PES-softening bias (~meV/atom) that fine-tuning roughly halves — trust the stage for *ranking* (where error largely cancels), DFT-spot-check before quoting absolute entropies; (2) for metals the **electronic** free energy (Fermi smearing, electron-phonon) is non-negligible at synthesis T and is **not** in $F_{\text{vib}}(T)$ — a future term, not captured here.

8 Status

- Classifiers (`_synth_classifiers.py`): implemented, unit-tested on a metallic Cu-Au hull (on-hull preferred polymorph → synthesizable; +25 meV/atom worse polymorph → maybe; +75 meV/atom → unlikely; off-hull Cu₂Au → correct decomposition products). Finite-T hull tested to flip a ranking (a higher-entropy CuAu polymorph overtakes the 0 K winner at 900 K).
- Workchain + DB write-back + MainWorkChain wiring + entry points: implemented.
- Phonon runner (`codes/files/phonon.py`) + `codes/phonon/` plugin: implemented; the runner end-to-end tested with EMT on fcc Cu (ZPE +0.033 eV/atom at 0 K → −0.51 eV/atom at 1200 K, thermal expansion, no imaginary modes).
- Open seams: trained PU model (`pu_model_path`); MLIP phonon fine-tuning for quantitative absolute free energies; an electronic free-energy term for metals.